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Miniature telephoto lens assembly

Abstract

An optical lens assembly includes five lens elements and provides a $TTL/EFL < 1.0$. In an embodiment, the focal length of the first lens element $f_1 < TTL/2$, an air gap between first and second lens elements is smaller than half the second lens element thickness, an air gap between the third and fourth lens elements is greater than $TTL/5$ and an air gap between the fourth and fifth lens elements is smaller than about 1.5 times the fifth lens element thickness. All lens elements may be aspheric.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/938,337 filed Nov. 6, 2024 (now allowed), which was a continuation of U.S. patent application Ser. No. 18/604,959 filed Mar. 14, 2024, now U.S. Pat. No. 12,169,266, which was a continuation of U.S. patent application Ser. No. 18/504,395 filed Nov. 8, 2023, now U.S. Pat. No. 11,953,659, which was a continuation of U.S. patent application Ser. No.

18/054,905 filed Nov. 13, 2022, now U.S. Pat. No. 11,835,694, which was a continuation of U.S. patent application Ser. No. 17/499,878 filed Oct. 13, 2021, now U.S. Pat. No. 12,072,475, which was a continuation of U.S. patent application Ser. No. 16/872,934 filed May 12, 2020, now abandoned, which was a continuation of U.S. patent application Ser. No. 16/829,804 filed Mar. 25, 2020, now U.S. Pat. No. 11,125,980, which was a continuation of U.S. patent application Ser. No. 16/665,977 filed Oct. 28, 2019, now U.S. Pat. No. 10,795,134, which was a continuation of U.S. patent application Ser. No. 16/296,272 filed Mar. 8, 2019, now U.S. Pat. No. 10,488,630, which was a continuation of U.S. patent application Ser. No. 15/976,391, now U.S. Pat. No. 10,330,897, and Ser. No. 15/976,422, now U.S. Pat. No. 10,317,647 filed May 10, 2018, which were a continuation of U.S. patent application Ser. No. 15/817,235 filed Nov. 19, 2017, now U.S. Pat. No. 10,324,277, which was a continuation of U.S. patent application Ser. No. 15/418,925 filed Jan. 30, 2017, now U.S. Pat. No. 9,857,568, which was a continuation in part of U.S. patent application Ser. No. 15/170,472 filed Jun. 1, 2016, now U.S. Pat. No. 9,568,712, which was a continuation of U.S. patent application Ser. No. 14/932,319 filed Nov. 4, 2015, now U.S. Pat. No. 9,402,032, which was a continuation of U.S. patent application Ser. No. 14/367,924 filed Sep. 19, 2014, now abandoned, which was a 371 of international application PCT/IB2014/062465 filed Jun. 20, 2014, and is related to and claims priority from U.S. Provisional Patent Application No. 61/842,987 filed Jul. 4, 2013, which is incorporated herein by reference in its entirety.

FIELD

(1) Embodiments disclosed herein relate to an optical lens system and lens assembly, and more particularly, to a miniature telephoto lens assembly included in such a system and used in a portable electronic product such as a cellphone.

BACKGROUND

(2) Digital camera modules are currently being incorporated into a variety of host devices. Such host devices include cellular telephones, personal data assistants (PDAs), computers, and so forth. Consumer demand for digital camera modules in host devices continues to grow. Cameras in cellphone devices in particular require a compact imaging lens system for good quality imaging and with a small total track length (TTL). Conventional lens assemblies comprising four lens elements are no longer sufficient for good quality imaging in such devices. The latest lens assembly designs, e.g. as in U.S. Pat. No. 8,395,851, use five lens elements. However, the design in U.S. Pat. No. 8,395,851 suffers from at least the fact that the TTL/EFL (effective focal length) ratio is too large.

(3) Therefore, a need exists in the art for a five lens element optical lens assembly that can provide a small TTL/EFL ratio and better image quality than existing lens assemblies.

SUMMARY

(4) Embodiments disclosed herein refer to an optical lens assembly comprising, in order from an object side to an image side: a first lens element with positive refractive power having a convex object-side surface, a second lens element with negative refractive power having a thickness $d_{sub.2}$ on an optical axis and separated from the first lens element by a first air gap, a third lens element with negative refractive power and separated from the second lens element by a second air gap, a fourth lens element having a positive refractive power and separated from the third lens element by a third air gap, and a fifth lens element having a negative refractive power, separated from the fourth lens element by a fourth air gap, the fifth lens element having a thickness $d_{sub.5}$ on the optical axis.

(5) An optical lens system incorporating the lens assembly may further include a stop positioned before the first lens element, a glass window disposed between the image-side surface of the fifth lens element and an image sensor with an image plane on which an image of the object is formed.

(6) The effective focal length of the lens assembly is marked "EFL" and the total track length on an optical axis between the object-side surface of the first lens element and the electronic sensor is

marked “TTL”. In all embodiments, TTL is smaller than the EFL, i.e. the TTL/EFL ratio is smaller than 1.0. In some embodiments, the TTL/EFL ratio is smaller than 0.9. In an embodiment, the TTL/EFL ratio is about 0.85. In all embodiments, the lens assembly has an F number $F \# < 3.2$. In an embodiment, the focal length of the first lens element f_1 is smaller than $TTL/2$, the first, third and fifth lens elements have each an Abbe number (“Vd”) greater than 50, the second and fourth lens elements have each an Abbe number smaller than 30, the first air gap is smaller than $d_{sub.2/2}$, the third air gap is greater than $TTL/5$ and the fourth air gap is smaller than $1.5d_{sub.5}$. In some embodiments, the surfaces of the lens elements may be aspheric.

(7) In an optical lens assembly disclosed herein, the first lens element with positive refractive power allows the TTL of the lens system to be favorably reduced. The combined design of the first, second and third lens elements plus the relative short distances between them enable a long EFL and a short TTL. The same combination, together with the high dispersion (low Vd) for the second lens element and low dispersion (high Vd) for the first and third lens elements, also helps to reduce chromatic aberration. In particular, the ratio $TTL/EFL < 1.0$ and minimal chromatic aberration are obtained by fulfilling the relationship $1.2 \times |f_3| > |f_2| > 1.5 \times f_1$, where “f” indicates the lens element effective focal length and the numerals 1, 2, 3, 4, 5 indicate the lens element number.

(8) The conditions $TTL/EFL < 1.0$ and $F \# < 3.2$ can lead to a large ratio $L11/L1e$ (e.g. larger than 4) between the largest width (thickness) $L11$ and the smallest width (thickness) of the first lens element (facing the object) $L1e$. The largest width is along the optical axis and the smallest width is of a flat circumferential edge of the lens element. $L11$ and $L1e$ are shown in each of elements **102**, **202** and **302**. A large $L11/L1e$ ratio (e.g. >4) impacts negatively the manufacturability of the lens and its quality. Advantageously, the present inventors have succeeded in designing the first lens element to have a $L11/L1e$ ratio smaller than 4, smaller than 3.5, smaller than 3.2, smaller than 3.1 (respectively 3.01 for element **102** and 3.08 for element **302**) and even smaller than 3.0 (2.916 for element **202**). The significant reduction in the $L11/L1e$ ratio improves the manufacturability and increases the quality of lens assemblies disclosed herein.

(9) The relatively large distance between the third and the fourth lens elements plus the combined design of the fourth and fifth lens elements assist in bringing all fields' focal points to the image plane. Also, because the fourth and fifth lens elements have different dispersions and have respectively positive and negative power, they help in minimizing chromatic aberration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A shows a first embodiment of an optical lens system disclosed herein;
- (2) FIG. 1B shows the modulus of the optical transfer function (MTF) vs. focus shift of the entire optical lens assembly for various fields in the first embodiment;
- (3) FIG. 1C shows the distortion vs. field angle (+Y direction) in percent in the first embodiment;
- (4) FIG. 2A shows a second embodiment of an optical lens system disclosed herein;
- (5) FIG. 2B shows the MTF vs. focus shift of the entire optical lens assembly for various fields in the second embodiment;
- (6) FIG. 2C shows the distortion +Y in percent in the second embodiment;
- (7) FIG. 3A shows a third embodiment of an optical lens system disclosed herein;
- (8) FIG. 3B shows the MTF vs. focus shift of the entire optical lens system for various fields in the third embodiment;
- (9) FIG. 3C shows the distortion +Y in percent in the third embodiment.

DETAILED DESCRIPTION

(10) In the following description, the shape (convex or concave) of a lens element surface is defined as viewed from the respective side (i.e. from an object side or from an image side). FIG. 1A

shows a first embodiment of an optical lens system disclosed herein and marked **100**. FIG. **1B** shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **100**. FIG. **1C** shows the distortion +Y in percent vs. field. Embodiment **100** comprises in order from an object side to an image side: an optional stop **101**; a first plastic lens element **102** with positive refractive power having a convex object-side surface **102a** and a convex or concave image-side surface **102b**; a second plastic lens element **104** with negative refractive power and having a meniscus convex object-side surface **104a**, with an image side surface marked **104b**; a third plastic lens element **106** with negative refractive power having a concave object-side surface **106a** with an inflection point and a concave image-side surface **106b**; a fourth plastic lens element **108** with positive refractive power having a positive meniscus, with a concave object-side surface marked **108a** and an image-side surface marked **108b**; and a fifth plastic lens element **110** with negative refractive power having a negative meniscus, with a concave object-side surface marked **110a** and an image-side surface marked **110b**. The optical lens system further comprises an optional glass window **112** disposed between the image-side surface **110b** of fifth lens element **110** and an image plane **114** for image formation of an object. Moreover, an image sensor (not shown) is disposed at image plane **114** for the image formation.

(11) In embodiment **100**, all lens element surfaces are aspheric. Detailed optical data is given in Table 1, and the aspheric surface data is given in Table 2, wherein the units of the radius of curvature (R), lens element thickness and/or distances between elements along the optical axis and diameter are expressed in mm. “Nd” is the refraction index. The equation of the aspheric surface profiles is expressed by:

$$(12) z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + {}_1r^2 + {}_2r^4 + {}_3r^6 + {}_4r^8 + {}_5r^{10} + {}_6r^{12} + {}_7r^{14}$$

where r is distance from (and perpendicular to) the optical axis, k is the conic coefficient, c=1/R where R is the radius of curvature, and a are coefficients given in Table 2. In the equation above as applied to embodiments of a lens assembly disclosed herein, coefficients α.sub.1 and α.sub.7 are zero. Note that the maximum value of r “max r”=Diameter/2. Also note that Table 1 (and in Tables 3 and 5 below), the distances between various elements (and/or surfaces) are marked “Lmn” (where m refers to the lens element number, n=1 refers to the element thickness and n=2 refers to the air gap to the next element) and are measured on the optical axis z, wherein the stop is at z=0. Each number is measured from the previous surface. Thus, the first distance −0.466 mm is measured from the stop to surface **102a**, the distance L11 from surface **102a** to surface **102b** (i.e. the thickness of first lens element **102**) is 0.894 mm, the gap L12 between surfaces **102b** and **104a** is 0.020 mm, the distance L21 between surfaces **104a** and **104b** (i.e. thickness d.sub.2 of second lens element **104**) is 0.246 mm, etc. Also, L21=d.sub.2 and L51=d.sub.5. L11 for lens element **102** is indicated in FIG. **1A**. Also indicated in FIG. **1A** is a width L1e of a flat circumferential edge (or surface) of lens element **102**. L11 and L1e are also indicated for each of first lens elements **202** and **302** in, respectively, embodiments **200** (FIG. **2A**) and **300** (FIG. **3A**).

(13) TABLE-US-00001 TABLE 1 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd
[mm] 1 Stop Infinite −0.466 2.4 2 L11 1.5800 0.894 1.5345/57.095 2.5 3 L12 −11.2003 0.020 2.4
4 L21 33.8670 0.246 1.63549/23.91 2.2 5 L22 3.2281 0.449 1.9 6 L31 −12.2843 0.290
1.5345/57.095 1.9 7 L32 7.7138 2.020 1.8 8 L41 −2.3755 0.597 1.63549/23.91 3.3 9 L42 −1.8801
0.068 3.6 10 L51 −1.8100 0.293 1.5345/57.095 3.9 11 L52 −5.2768 0.617 4.3 12 Window Infinite
0.210 1.5168/64.17 3.0 13 Infinite 0.200 3.0

(14) TABLE-US-00002 TABLE 2 Conic coefficient # k α.sub.2 α.sub.3 α.sub.4 α.sub.5 α.sub.6 2
−0.4668 7.9218E−03 2.3146E−02 −3.3436E−02 2.3650E−02 −9.2437E−03 3 −9.8525
2.0102E−02 2.0647E−04 7.4394E−03 −1.7529E−02 4.5206E−03 4 10.7569 −1.9248E−03
8.6003E−02 1.1676E−02 −4.0607E−02 1.3545E−02 5 1.4395 5.1029E−03 2.4578E−01
−1.7734E−01 2.9848E−01 −1.3320E−01 6 0.0000 2.1629E−01 4.0134E−02 1.3615E−02
2.5914E−03 −1.2292E−02 7 −9.8953 2.3297E−01 8.2917E−02 −1.2725E−01 1.5691E−01

-5.9624E-02 8 0.9938 -1.3522E-02 -7.0395E-03 1.4569E-02 -1.5336E-02 4.3707E-03 9
-6.8097 -1.0654E-01 1.2933E-02 2.9548E-04 -1.8317E-03 5.0111E-04 10 -7.3161
-1.8636E-01 8.3105E-02 -1.8632E-02 2.4012E-03 -1.2816E-04 11 0.0000 -1.1927E-01
7.0245E-02 -2.0735E-02 2.6418E-03 -1.1576E-04

Embodiment **100** provides a field of view (FOV) of 44 degrees, with EFL=6.90 mm, F #=2.80 and TTL of 5.904 mm. Thus and advantageously, the ratio TTL/EFL=0.855. Advantageously, the Abbe number of the first, third and fifth lens element is 57.095. Advantageously, the first air gap between lens elements **102** and **104** (the gap between surfaces **102b** and **104a**) has a thickness (0.020 mm) which is less than a tenth of thickness d.sub.2 (0.246 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, the third air gap between lens elements **106** and **108** has a thickness (2.020 mm) greater than TTL/5 (5.904/5 mm). Advantageously, the fourth air gap between lens elements **108** and **110** has a thickness (0.068 mm) which is smaller than 1.5d.sub.5 (0.4395 mm).

(15) The focal length (in mm) of each lens element in embodiment **100** is as follows: f1=2.645, f2=-5.578, f3=-8.784, f4=9.550 and f5=-5.290. The condition $1.2 \times |f3| > |f2| < 1.5 \times f1$ is clearly satisfied, as $1.2 \times 8.787 > 5.578 > 1.5 \times 2.645$. f1 also fulfills the condition $f1 < TTL/2$, as $2.645 < 2.952$.

(16) Using the data from row #2 in Tables 1 and 2, L1e in lens element **102** equals 0.297 mm, yielding a center-to-edge thickness ratio L11/L1e of 3.01.

(17) FIG. 2A shows a second embodiment of an optical lens system disclosed herein and marked **200**. FIG. 2B shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **200**. FIG. 2C shows the distortion +Y in percent vs. field. Embodiment **200** comprises in order from an object side to an image side: an optional stop **201**; a first plastic lens element **202** with positive refractive power having a convex object-side surface **202a** and a convex or concave image-side surface **202b**; a second glass lens element **204** with negative refractive power, having a meniscus convex object-side surface **204a**, with an image side surface marked **204b**; a third plastic lens element **206** with negative refractive power having a concave object-side surface **206a** with an inflection point and a concave image-side surface **206b**; a fourth plastic lens element **208** with positive refractive power having a positive meniscus, with a concave object-side surface marked **208a** and an image-side surface marked **208b**; and a fifth plastic lens element **210** with negative refractive power having a negative meniscus, with a concave object-side surface marked **110a** and an image-side surface marked **210b**. The optical lens system further comprises an optional glass window **212** disposed between the image-side surface **210b** of fifth lens element **210** and an image plane **214** for image formation of an object.

(18) In embodiment **200**, all lens element surfaces are aspheric. Detailed optical data is given in Table 3, and the aspheric surface data is given in Table 4, wherein the markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for embodiment **100**.

(19) TABLE-US-00003 TABLE 3 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd [mm]
1 Stop Infinite -0.592 2.5 2 L11 1.5457 0.898 1.53463/56.18 2.6 3 L12 -127.7249 0.129 2.6 4 L21 6.6065 0.251 1.91266/20.65 2.1 5 L22 2.8090 0.443 1.8 6 L31 9.6183 0.293 1.53463/56.18 1.8 7 L32 3.4694 1.766 1.7 8 L41 -2.6432 0.696 1.632445/23.35 3.2 9 L42 -1.8663 0.106 3.6 10 L51 -1.4933 0.330 1.53463/56.18 3.9 11 L52 -4.1588 0.649 4.3 12 Window Infinite 0.210 1.5168/64.17 5.4 13 Infinite 0.130 5.5

(20) TABLE-US-00004 TABLE 4 Conic coefficient # k $\alpha_{sub.2}$ $\alpha_{sub.3}$ $\alpha_{sub.4}$ $\alpha_{sub.5}$ $\alpha_{sub.6}$ 2
0.0000 -2.7367E-03 2.8779E-04 -4.3661E-03 3.0069E-03 -1.2282E-03 3 -10.0119 4.0790E-02 -1.8379E-02 2.2562E-02 -1.7706E-02 4.9640E-03 4 10.0220 4.6151E-02 5.8320E-02 -2.0919E-02 -1.2846E-02 8.8283E-03 5 7.2902 3.6028E-02 1.1436E-01 -1.9022E-02 4.7992E-03 -3.4079E-03 6 0.0000 1.6639E-01 5.6754E-02 -1.2238E-02 -1.8648E-02 1.9292E-02 7 8.1261 1.5353E-01 8.1427E-02 -1.5773E-01 1.5303E-01 -4.6064E-02 8 0.0000 -3.2628E-02 1.9535E-02 -1.6716E-02 -2.0132E-03 2.0112E-03 9

0.000 1.5173E-02 -1.2252E-02 3.3611E-03 -2.5303E-03 8.4038E-04 10 -4.7688
-1.4736E-01 7.6335E-02 -2.5539E-02 5.5897E-03 -5.0290E-04 11 0.00E+00 -8.3741E-02
4.2660E-02 -8.4866E-03 1.2183E-04 7.2785E-05

(21) Embodiment **200** provides a FOV of 43.48 degrees, with EFL=7 mm, F #=2.86 and TTL=5.90 mm. Thus and advantageously, the ratio TTL/EFL=0.843. Advantageously, the Abbe number of the first, third and fifth lens elements is 56.18. The first air gap between lens elements **202** and **204** has a thickness (0.129 mm) which is about half the thickness d.sub.2 (0.251 mm). Advantageously, the Abbe number of the second lens element is 20.65 and of the fourth lens element is 23.35.

Advantageously, the third air gap between lens elements **206** and **208** has a thickness (1.766 mm) greater than TTL/5 (5.904/5 mm). Advantageously, the fourth air gap between lens elements **208** and **210** has a thickness (0.106 mm) which is less than 1.5×d.sub.5 (0.495 mm).

(22) The focal length (in mm) of each lens element in embodiment **200** is as follows: f1=2.851, f2=-5.468, f3=-10.279, f4=7.368 and f5=-4.536. The condition $1.2 \times |f3| > |f2| < 1.5 \times f1$ is clearly satisfied, as $1.2 \times 10.279 > 5.468 > 1.5 \times 2.851$. f1 also fulfills the condition $f1 < TTL/2$, as $2.851 < 2.950$.

(23) Using the data from row #2 in Tables 3 and 4, L1e in lens element **202** equals 0.308 mm, yielding a center-to-edge thickness ratio L11/L1e of 2.916.

(24) FIG. 3A shows a third embodiment of an optical lens system disclosed herein and marked **300**. FIG. 3B shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **300**. FIG. 3C shows the distortion +Y in percent vs. field. Embodiment **300** comprises in order from an object side to an image side: an optional stop **301**; a first glass lens element **302** with positive refractive power having a convex object-side surface **302a** and a convex or concave image-side surface **302b**; a second plastic lens element **204** with negative refractive power, having a meniscus convex object-side surface **304a**, with an image side surface marked **304b**; a third plastic lens element **306** with negative refractive power having a concave object-side surface **306a** with an inflection point and a concave image-side surface **306b**; a fourth plastic lens element **308** with positive refractive power having a positive meniscus, with a concave object-side surface marked **308a** and an image-side surface marked **308b**; and a fifth plastic lens element **310** with negative refractive power having a negative meniscus, with a concave object-side surface marked **310a** and an image-side surface marked **310b**. The optical lens system further comprises an optional glass window **312** disposed between the image-side surface **310b** of fifth lens element **310** and an image plane **314** for image formation of an object.

(25) In embodiment **300**, all lens element surfaces are aspheric. Detailed optical data is given in Table 5, and the aspheric surface data is given in Table 6, wherein the markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for embodiments **100** and **200**.

(26) TABLE-US-00005 TABLE 5 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd
[mm] 1 Stop Infinite -0.38 2.4 2 L11 1.5127 0.919 1.5148/63.1 2.5 3 L12 -13.3831 0.029 2.3 4
L21 8.4411 0.254 1.63549/23.91 2.1 5 L22 2.6181 0.426 1.8 6 L31 -17.9618 0.265
1.5345/57.09 1.8 7 L32 4.5841 1.998 1.7 8 L41 -2.8827 0.514 1.63549/23.91 3.4 9 L42
-1.9771 0.121 3.7 10 L51 -1.8665 0.431 1.5345/57.09 4.0 11 L52 -6.3670 0.538 4.4 12 Window
Infinite 0.210 1.5168/64.17 3.0 13 Infinite 0.200 3.0

(27) TABLE-US-00006 TABLE 6 Conic coefficient # k $\alpha_{sub.2}$ $\alpha_{sub.3}$ $\alpha_{sub.4}$ $\alpha_{sub.5}$ $\alpha_{sub.6}$ 2
-0.534 1.3253E-02 2.3699E-02 -2.8501E-02 1.7853E-02 -4.0314E-03 3 -13.473 3.0077E-02
4.7972E-03 1.4475E-02 -1.8490E-02 4.3565E-03 4 -10.132 7.0372E-04 1.1328E-01
1.2346E-03 -4.2655E-02 8.8625E-03 5 5.180 -1.9210E-03 2.3799E-01 -8.8055E-02
2.1447E-01 -1.2702E-01 6 0.000 2.6780E-01 1.8129E-02 -1.7323E-02 3.7372E-02
-2.1356E-02 7 10.037 2.7660E-01 -1.0291E-02 -6.0955E-02 7.5235E-02 -1.6521E-02 8
1.703 2.6462E-02 -1.2633E-02 -4.7724E-04 -3.2762E-03 1.6551E-03 9 -1.456 5.7704E-03
-1.8826E-02 5.1593E-03 -2.9999E-03 8.0685E-04 10 -6.511 -2.1699E-01 1.3692E-01
-4.2629E-02 6.8371E-03 -4.1415E-04 11 0.000 -1.5120E-01 8.6614E-02 -2.3324E-02

2.7361E-03 - 1.1236E-04

(28) Embodiment **300** provides a FOV of 44 degrees, EFL=6.84 mm, F #=2.80 and TTL=5.904 mm. Thus and advantageously, the ratio TTL/EFL=0.863. Advantageously, the Abbe number of the first lens element is 63.1, and of the third and fifth lens elements is 57.09. The first air gap between lens elements **302** and **304** has a thickness (0.029 mm) which is about 1/10th the thickness d.sub.2 (0.254 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, the third air gap between lens elements **306** and **308** has a thickness (1.998 mm) greater than TTL/5 (5.904/5 mm). Advantageously, the fourth air gap between lens elements **208** and **210** has a thickness (0.121 mm) which is less than 1.5d.sub.5 (0.6465 mm).

(29) The focal length (in mm) of each lens element in embodiment **300** is as follows: f1=2.687, f2=-6.016, f3=-6.777, f4=8.026 and f5=-5.090. The condition $1.2 \times |f3| > |f2| < 1.5 \times f1$ is clearly satisfied, as $1.2 \times 6.777 > 6.016 > 1.5 \times 2.687$. f1 also fulfills the condition $f1 < TTL/2$, as $2.687 < 2.952$.

(30) Using the data from row #2 in Tables 5 and 6, L1e in lens element **302** equals 0.298 mm, yielding a center-to-edge thickness ratio L11/L1e of 3.08.

(31) While this disclosure has been described in terms of certain embodiments and generally associated methods, alterations and permutations of the embodiments and methods will be apparent to those skilled in the art. The disclosure is to be understood as not limited by the specific embodiments described herein, but only by the scope of the appended claims.

Claims

1. A lens system, comprising: a lens assembly that includes a plurality of refractive lens elements arranged along an optical axis and including, in order from an object side to an image side, a first lens group comprising at least three lens elements, and a second lens group comprising at least two lens elements, the lens assembly having an effective focal length EFL and a total track length TTL; a stop positioned before the first lens element of the first lens group; and a window positioned between the plurality of refractive lens elements and an image plane, wherein all the lens elements are made of plastic, wherein a largest gap between lens elements d.sub.MAX is a gap between the first and second lens groups, wherein at least one surface of at least one of the plurality of refractive lens elements is aspheric, wherein $TTL/EFL < 0.9$, $d.sub.MAX/TTL > 0.2$ and a f number f # < 2.9, wherein a center thickness of each lens element is equal to or larger than 0.2 mm, and wherein the lens system is included in a mobile device.

2. The lens system of claim 1, wherein d.sub.MAX is larger than 2.25 times any other gap between lens elements.

3. The lens system of claim 1, wherein d.sub.MAX is larger than 2.5 times any other gap between lens elements.

4. The lens system of claim 1, wherein d.sub.MAX is larger than 2.75 times any other gap between lens elements.

5. The lens system of claim 1, wherein a smallest gap between lens elements is d.sub.MIN, wherein $d.sub.MIN < 0.25$ mm.

6. The lens system of claim 5, wherein $d.sub.MIN < 0.2$ mm.

7. The lens system of claim 5, wherein $d.sub.MIN < 0.15$ mm.

8. The lens system of claim 5, wherein a ratio $d.sub.MIN/d.sub.MAX < 0.2$.

9. The lens system of claim 5, wherein a ratio $d.sub.MIN/d.sub.MAX < 0.15$.

10. The lens system of claim 5, wherein $d.sub.MIN < 0.2$ mm, and wherein a ratio $d.sub.MIN/d.sub.MAX < 0.2$.

11. The lens system of claim 5, wherein $d.sub.MIN < 0.15$ mm, and wherein a ratio $d.sub.MIN/d.sub.MAX < 0.15$.

12. The lens system of claim 1, wherein the lens assembly includes five refractive lens elements.

13. The lens system of claim 1, wherein f # ≥ 2.8 .

14. The lens system of claim 1, wherein a ratio $d_{\text{sub.MAX}}/\text{TTL} > 0.25$.
 15. The lens system of claim 1, wherein a ratio $d_{\text{sub.MAX}}/\text{TTL} > 0.3$.
 16. The lens system of claim 1, wherein the first lens element of the first lens group has positive refractive power and the second lens element of the first lens group has negative refractive power.
 17. The lens system of claim 1, wherein the second lens group includes two adjacent lens elements having opposite refractive powers.
 18. The lens system of claim 1, wherein a focal length f_1 of the first lens element of the first lens group is smaller than $\text{TTL}/2$.
 19. The lens system of claim 1, wherein all lens element surfaces are aspheric.
 20. The lens system of claim 1, wherein the mobile device is a smartphone.
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